Accelerated Math II/III Name_____

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HW #8 - Solving Polynomial Equations by Graphing Date_____ Period____

Sketch the graph of each function. Use your graphing calculator to approximate each real zero to the nearest tenth.

1) $f(x) = x^3 - x^2 - 2$

2) $f(x) = x^4 - x^3 - 2x^2 - 1$

3) $f(x) = x^2 + 6x + 3$

4) $f(x) = -x^3 - 8x^2 - 16x - 7$

5) $f(x) = x^3 - 3x^2 + 6$

6) $f(x) = -x^5 + 4x^3 - x - 2$

7) $f(x) = x^4 - 4x^3 + 4x^2 + 3$

8) $f(x) = x^5 - 4x^3 + 2x - 3$

Factor each completely.

9) $27x^3 - 125$

10) $24x^3 - 3$

Find all roots. Write complex solutions in standard form ($a + ib$).

11) $3x^3 - x^2 - 2x = 0$

12) $3x^3 + 8x^2 + 5x = 0$

13) $5x^4 - 12x^2 - 32 = 0$

14) $2x^4 + 25x^2 + 63 = 0$

15) $5x^4 + 3x^3 - 15x^2 - 9x = 0$

16) $5x^4 + 3x^3 - 5x^2 - 3x = 0$

17) $x^4 + 125x = 0$

18) $8x^4 - x = 0$

Solve each equation by completing the square.

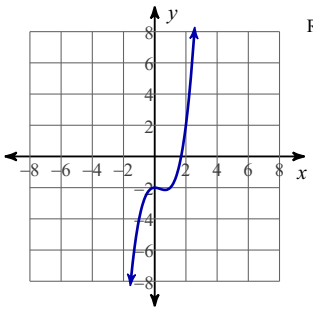
19) $n^2 - 10n - 40 = 5$

20) $8x^2 - 16x - 46 = 3$

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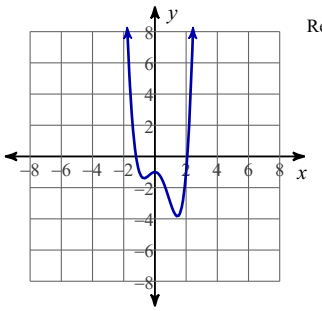
Answers to HW #8 - Solving Polynomial Equations by Graphing

1)



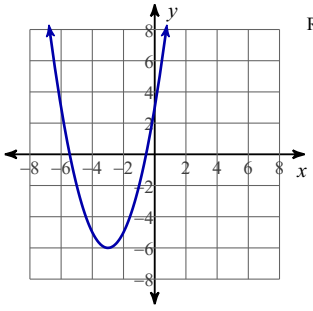
Real Zeros: 1.7

2)



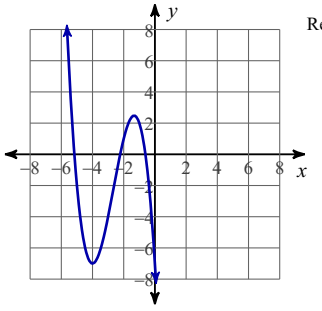
Real Zeros: −1.2, 2.1

3)



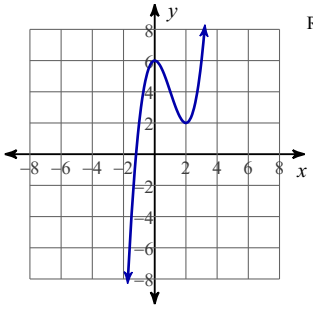
Real Zeros: −5.4, −0.6

4)



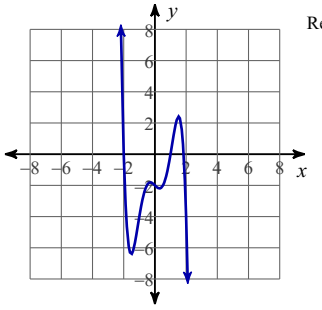
Real Zeros: −5.2, −2.2, −0.6

5)



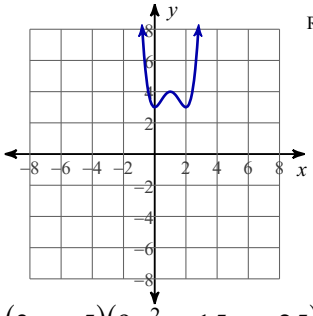
Real Zeros: −1.2

6)



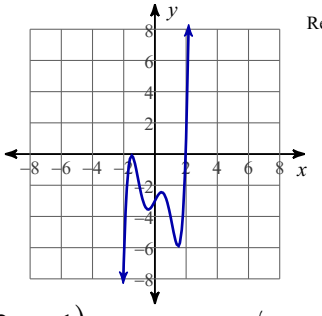
Real Zeros: −2, 1, 1.8

7)



Real Zeros: None

8)



Real Zeros: 2

9) $(3x-5)(9x^2+15x+25)$

10) $3(2x-1)(4x^2+2x+1)$

11) $\left\{0, -\frac{2}{3}, 1\right\}$

12) $\left\{0, -\frac{5}{3}, -1\right\}$

13) $\left\{\frac{2i\sqrt{10}}{5}, -\frac{2i\sqrt{10}}{5}, 2, -2\right\}$

14) $\left\{\frac{i\sqrt{14}}{2}, -\frac{i\sqrt{14}}{2}, 3i, -3i\right\}$

15) $\left\{0, -\frac{3}{5}, \sqrt{3}, -\sqrt{3}\right\}$

16) $\left\{0, -\frac{3}{5}, 1, -1\right\}$

17) $\left\{0, -5, \frac{5+5i\sqrt{3}}{2}, \frac{5-5i\sqrt{3}}{2}\right\}$

18) $\left\{0, \frac{1}{2}, \frac{-1+i\sqrt{3}}{4}, \frac{-1-i\sqrt{3}}{4}\right\}$

19) $\left\{5+\sqrt{70}, 5-\sqrt{70}\right\}$

20) $\left\{\frac{4+\sqrt{114}}{4}, \frac{4-\sqrt{114}}{4}\right\}$

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